

Title of the thesis:

Prague Mapped by People and Satellites

Assignment of the thesis:

The aim of the thesis is to create an interactive and user-friendly visualization of an open dataset from participatory emotional mapping and to explore possibilities for integrating Copernicus Earth Observation data into the spatial analysis and visualization process.

The dataset, collected in Prague in 2021 via the EmotionalMaps.eu platform, contains over 98,000 geolocated points and more than 30,000 citizen comments. The student will preprocess and analyze the dataset to uncover spatial patterns and trends that are not immediately visible, using appropriate GIS and statistical methods. A key part of the work will be the exploration of how Copernicus data—such as Sentinel-2 imagery, urban heat islands, or land cover information—can enhance the understanding of these perceptions, particularly in relation to environmental quality and urban resilience.

Based on the findings, the student will design and implement an interactive web-based map or application that visualizes both the participatory and satellite data in a coherent, comprehensible, and visually engaging way. The application should allow filtering by theme, location, or demographic characteristics and should integrate textual comments from respondents.

An important component of the work will be the evaluation of the user interface through usability testing, ideally involving both lay users and professionals from urban planning. The outcome of the thesis will be a fully functional interactive visualization.

The student will create a website about the thesis following the rules available on the department's website and a poster about the diploma thesis in A2 format. The student will submit the entire text (text, attachments, poster, outputs, input, and output data) in digital form on a storage medium and the text of the thesis in two bound copies to the secretary of the department.

The extent of graphic works:

as required

The extent of the thesis:

maximum 40 pages of text excluding appendices

List of recommended literature:

Pánek, J., Barvíř, R., Koniček, J., & Brlík, M. (2021). The emotional map of Prague—data on what locals think about the Czech capital?. *Data in brief*, 39, 107649.

Pánek, J. (2011, September). Participatory and Public Participation GIS: A Phenomenon of Neocartography with a High Potential in Developing Countries. In 5th International Conference on Globalization, Czech Republic.

Květoňová, V., Pánek, J., Geletič, J., Šimáček, P., & Lehnert, M. (2024). Where is the heat threat in a city? Different perspectives on people-oriented and remote sensing methods: The case of Prague. *Heliyon*, 10(16).

1. The emotional map of Prague -data on what locals think about the Czech capital?

The study employed a **participatory crowdsourcing methodology** where citizens could mark locations and provide comments about their emotional perceptions of different areas in Prague. The platform allowed for both quantitative data (point locations) and qualitative data (user comments) to be collected simultaneously J. Pánek et al., 2021.

Based on the provided sources, the focus of the paper by Pánek et al. (2021) is **participatory emotional mapping to understand local perceptions of Prague**.

Main Research Focus: The paper presents data from participatory emotional mapping in Prague, Czech Republic, aimed at capturing what local residents think and feel about their capital city J. Pánek et al., 2021.

Key Objectives:

- **Data Collection:** The study collected comprehensive data including 98,364 points complemented with 30,941 comments from 5,973 respondents across Prague J. Pánek et al., 2021
- **Citizen Participation:** The research utilized a participatory approach where locals could directly contribute their perceptions and emotions about different areas of the city
- **Comprehensive Coverage:** The study covered the entire city of Prague (population 1,335,084) through their custom platform EmotionalMaps.eu J. Pánek et al., 2021

Methodological Approach: The research employed a structured data collection system with eight common questions/statements applied across all of Prague, while allowing each of the 27 Prague districts to add up to seven additional location-specific questions J. Pánek et al., 2021.

2.Mapping citizens' emotions: participatory planning support system in Olomouc, Czech Republic.

- To present a case study of using a web-based participatory mapping platform (the “Emotional Maps” tool) in the city of Olomouc, Czech Republic, where citizens marked locations and provided comments about their urban environment. [IDEAS/RePEc](#)
- To support urban planning by collecting spatial data on citizens' perceptions (comfort, safety, attractiveness, public transport satisfaction, suggestions for future development) through this mapping.

2.Participatory and Public Participation GIS: A Phenomenon of Neocartography with a High Potential in Developing Countries. In 5th International Conference on Globalization, Czech Republic.

The main goal of this paper was to **introduce and discuss the concept of Participatory GIS (PGIS) and Public Participation GIS (PPGIS)** as new approaches in **neocartography** — that is, modern, digital, and interactive mapping.

Pánek aimed to show how **ordinary citizens**, not just experts, can take part in **creating and using spatial data** to express their opinions, experiences, and local knowledge.

He emphasized that these methods have **great potential for developing countries**, where local communities can use participatory mapping to **influence planning, resource management, and decision-making**, even with limited resources or formal GIS training.

- **Discussed** the opportunities and challenges (such as data quality, accessibility, and digital literacy) in applying these approaches, especially in developing contexts.

3.Where is the heat threat in a city? Different perspectives on people-oriented and remote sensing methods: The case of Prague.

The main goal of this study was to **compare two different ways of identifying heat risk in a city** —

1. **People-oriented methods** (e.g., citizens' perceptions of heat discomfort gathered through participatory mapping or surveys), and
2. **Remote sensing methods** (e.g., satellite-derived land surface temperature and urban heat island data).

The researchers focused on **Prague, Czech Republic**, and aimed to see **how well citizens' subjective experiences of heat matched the objective data** from satellite imagery. By combining and comparing both perspectives, the study sought to **understand urban heat risk more comprehensively**, showing that remote sensing alone cannot fully explain how people actually feel and experience heat in their daily environments.

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4. Social networking site photos reveal significant emotional variations across urban green spaces in northern China, with landscape characteristics strongly influencing visitor emotions.

Focus: Using geotagged photos from social media to assess emotional responses to urban green spaces.

Researchers collected thousands of visitor photos across multiple studies: Ping Liu et al., 2021 analyzed 2,031 facial photos from 18 forest parks, finding that increased green space and intact park areas correlated with higher happiness frequencies. Jie Zhang et al., 2021 similarly examined 1,145 photos across 24 parks, discovering that green and blue spaces contribute differently to emotional experiences, with green spaces positively impacting happy scores and blue spaces reducing sad scores.

Demographic variations were notable: male adolescents smiled more frequently, and females expressed more positive emotions overall. Jie Zhang et al., 2021. The studies demonstrate a robust methodology of using social media facial recognition to quantify emotional responses to urban green spaces, providing nuanced insights into how park design and landscape metrics influence visitor well-being.

The methodology for quantifying emotional differences in urban green spaces from social networking site photos involves several key components:

Data Collection: Ping Liu et al., 2021 collected facial photos of 2,031 people from Sina Weibo across 18 forest parks in Northern China in 2020. Jie Zhang et al., 2021 gathered 1,145 photos and selfies from Sina Weibo (~50 individuals per park) across 24 parks in Northeast China. Hongxu Wei et al., 2019 obtained 935 photos with location records from Sina Micro-Blog during China's Golden Week holidays.

Facial Expression Analysis: Multiple studies used FireFACE software (v1.0) to recognize and rate facial expressions Jie Zhang et al., 2021 & 1 others. Hongxu Wei et al., 2019 used FireFace software to assess scores for neutral, happy, sad, angry, surprised, scared, disgusted, and contempt expressions. Emotions were typically categorized as happy or sad scores and calculated as positive response index (PRI; happy-sad).

Landscape Assessment: Ping Liu et al., 2021 used satellite imagery analysis to assess elevation and pattern sizes of green and blue space areas. Siying Huang et al., 2022 employed semantic segmentation through machine learning to analyze panoramic photos and assess the Panoramic Green View Index (PGVI).

Statistical Analysis: Methods included multivariate linear regression Jie Zhang et al., 2021, Friedman's test, correlation analysis, and Poisson regression Hongxu Wei et al., 2019, and split-plot analysis of variance Siying Huang et al., 2022.

Methodology: Exploring the Relationship Between Citizens' Emotions and Urban Green Spaces

1. Overview

This methodology outlines the steps to analyze how urban green spaces influence citizens' emotional perceptions in Prague. The process combines participatory emotional mapping data with Copernicus Earth Observation datasets to identify spatial relationships between environmental quality and public sentiment.

2. Data Source

The study uses multiple open-source datasets to integrate emotional and environmental information. The core dataset is the **EmotionalMaps.eu dataset** (Prague, 2021), which includes 98,364 geolocated points and over 30,000 citizen comments representing public emotions across the city. To assess environmental conditions, **Sentinel-2 imagery** from the Copernicus Open Access Hub provides 10 m multispectral data for deriving vegetation indices such as NDVI, which reflect green-space quality. Additionally, **Copernicus Urban Atlas** and **CORINE Land Cover** datasets offer detailed land-use and land-cover classifications to identify urban green areas and surface types. Finally, **OpenStreetMap (OSM)** data are used to validate and refine green-space boundaries, adding contextual information such as parks, roads, and water bodies. Together, these datasets enable a comprehensive spatial analysis linking emotional perceptions with urban environmental characteristics.

3. Data Preprocessing

1. Emotional Data Cleaning:

- Remove duplicates, outliers, and incomplete records.
- Normalize coordinate reference systems (WGS84 → UTM Zone 33N).
- Categorize emotions as *positive*, *negative*, or *neutral* based on keywords or pre-coded attributes.

2. Sentinel-2 Image Processing:

- Acquire cloud-free images for 2021 (matching emotional data period).
- Perform atmospheric correction using **Sentinel Application Platform (SNAP)**.
- Calculate **Normalized Difference Vegetation Index (NDVI)**:

$$NDVI = \frac{NIR - RED}{NIR + RED}$$

- Resample NDVI to 10 m resolution and mask non-urban or water areas.

3. Green Space Extraction:

- Derive urban green-space polygons from **NDVI thresholds** (e.g., $NDVI > 0.3$).
- Combine with **Urban Atlas / OSM park layers** for validation and refinement.

“For my thesis, I'd like to focus on exploring how citizens' emotional perceptions in Prague—collected through the EmotionalMaps.eu dataset—relate to the quality and distribution of urban green spaces. The dataset includes almost 100,000 geolocated points and over 30,000 comments, which I'll preprocess and classify by emotional tone, such as positive or negative.

My main goal is to integrate this participatory emotional data with Copernicus Earth Observation data, especially Sentinel-2 imagery, to analyze how environmental characteristics influence people's emotions. For instance, I can calculate NDVI values or identify green-space accessibility around each emotional point to see whether greener or cooler areas are linked to more positive emotions.

So essentially, I want to connect **how people feel** with **what the satellite sees**, focusing on green spaces as a key factor shaping emotional well-being in the city.”